

SURBHI DHIMAN

surbhidhiman@live.com ◇ [linkedin.com/surbhidhiman](https://www.linkedin.com/surbhidhiman) ◇ github.com/surbhidhiman ◇ surbhidhiman.github.io

SUMMARY

I build end-to-end modeling and analytical systems that unite machine learning, optimization, and scientific reasoning. My work has covered supply chain optimization, probabilistic forecasting, NLP pipelines, and large-scale analytical workflows. I also have experience integrating heterogeneous data sources, implementing production-quality infrastructure, and translating real-world constraints into formal quantitative models. I focus on developing reliable, interpretable systems grounded in both mathematical rigor and practical deployment requirements.

Across commercial and government-scale environments, I have owned complex modeling problems from initial formulation through implementation, validation, and delivery. My projects include developing optimization engines, forecasting frameworks, and model-driven applications that support high-stakes operational decisions. I emphasize reproducibility, clarity of assumptions, and scalable design. My long-term trajectory centers on technical leadership in modeling and ML systems, building toward roles that direct modeling strategy, guide teams, and help shape the technical roadmap for scientifically grounded AI.

TECHNICAL EXPERTISE

Programming Languages	Python, SQL, Julia, C++, FORTRAN
Machine Learning and Modeling	Scientific machine learning, timeseries forecasting, probabilistic modeling, optimization-based modeling, supervised learning, interpretable ML
Numerical and Scientific Computing	Numerical ODE and PDE solvers, simulation-based modeling, computational physics, statistical analysis of physical systems, numerical stability and validation
Optimization and Forecasting	Mathematical optimization, inventory and supply chain optimization, scenario analysis, backtesting, risk and uncertainty quantification
Data and Systems	Relational and NoSQL databases, database schema design, ETL pipelines, large-scale data processing, data validation, production analytics systems
Scientific Software and Tooling	Python scientific stack, Julia scientific computing ecosystem, TensorFlow, scikit-learn, object-oriented design, version control (Git), reproducible research workflows

PROFESSIONAL EXPERIENCE

Data Scientist

Apr 2023 – Present

Booz Allen Hamilton

- Contribute to the development of a unified analytics and decision-support platform by designing large-scale ETL pipelines, analytical workflows, and machine-learning components that enable organization-wide data-driven decision making.
- Lead cross-functional initiatives across metric computation for congressional reporting, grant analytics applications, and scholarship ROI evaluations—owning data integration, validation, and quantitative modeling across heterogeneous datasets.
- Designed and implemented a probabilistic forecasting system in Python to model future metric trajectories and quantify uncertainty relative to dynamic performance thresholds.
- Developed advanced analytical tools including NLP pipelines to extract action items from federal budget documents, and cost-optimization analyses that reduced operational expenses across Databricks and Qlik-based systems.
- Modernized internal analytical infrastructure by refactoring dashboards, improving transparency of model inputs, assumptions, and metric lineage for both technical and non-technical stakeholders.

- Built optimization systems for supply chain decision support, including engines for replenishment recommendations, cash-flow maximization, and combinatorial container-loading to reduce waste and improve operational efficiency.
- Developed quantitative modeling tools such as a demand-forecasting web application for a Fortune 500 client and clustering algorithms for product velocity and profit-margin segmentation.
- Created modular components for product-location-level replenishment modeling to predict stock levels, purchase orders, lost sales, and financial impacts.
- Designed and maintained production grade software systems, including REST APIs, platform services, and ETL pipelines using Python, Flask, SQLAlchemy, PostgreSQL, MongoDB, and Kubernetes.
- Collaborated on a major database schema redesign and implemented CI/CD workflows with GitHub Actions to improve reliability, scalability, and development.
- Supported client onboarding by translating custom operational requirements into technical specifications and platform features.
- Built internal tools including a supply chain health dashboard and implemented user authentication and access control functionality using OAuth.

RESEARCH EXPERIENCE

Graduate Researcher

San Diego State University

Aug 2025 – Present

San Diego, CA

- Develop computational models for a unified research project investigating neutron star environments interactions with axions. This work combines physically motivated mathematical formulations with numerical solvers and machine learning based techniques to explore system dynamics.
- Formulate, implement, and validate simulation components using graduate-level training in computational physics, numerical methods, and scientific modeling, ensuring stability, interpretability, and physical consistency of results.

Undergraduate Researcher

Lawrence Berkeley National Laboratory

Sep 2016 – Jun 2017

Berkeley, CA

- Statistically measured the dimensions of large scale galaxy filaments and analyzed these results using Python as a cosmological probe under the supervision of the head scientist of the Cosmology Department.

PROJECTS

Probabilistic Time Series Forecasting Platform

Industry Project

Led development of a Python-based forecasting system to predict future business metrics and quantify uncertainty around target attainment. Designed end-to-end pipelines for data ingestion, validation, modeling, and probabilistic risk estimation used by senior stakeholders.

Optimization Models for Inventory Replenishment

Industry Project

Designed and implemented optimization models for inventory replenishment under real world operational constraints. Translated business objectives into formal objective functions and integrated recommendations into production systems.

Scientific Machine Learning for Physical Systems

Graduate Research (In Progress)

Developing computational models that combine classical numerical solvers and physics informed neural networks. Focuses on physically motivated formulations, numerical stability, and validation using Python and Julia.

Air Quality Forecasting and Public Health Analysis

Applied Research / Incubator Program

Contributed to a collaborative research project forecasting air pollutant concentrations and analyzing relationships

with public health outcomes using California datasets. Implemented classical and deep learning time series models (SARIMA, LSTM) and supported causal analysis.

Computational Astrophysics Modeling

Academic Research

Built statistical and computational models to analyze large-scale cosmic filaments as part of undergraduate research at Lawrence Berkeley National Laboratory, connecting physical theory, mathematical formulation, and Python-based data analysis.

EDUCATION

MS Computational Science, San Diego State University

Expected 2027

Focus: scientific computing, numerical methods, ML-based physical modeling

Coursework: Databases, Computational Methods, General Relativity

BA Astrophysics, Applied Mathematics, University of California, Berkeley

2015 - 2019

Concentration: Quantum Mechanics

OPEN SOURCE CONTRIBUTIONS

Open-Source Scientific Computing Contribution. [Oceananigans.jl](#)

Contributed to a numerical simulation codebase in the Julia ecosystem, implementing and testing scientific modeling components to improve documentation of the simulation framework.

LEADERSHIP

Data Science Team Lead, Forecasting Initiative. Led a small team in developing a Python-based forecasting and probabilistic risk modeling product for enterprise-level metrics, directing model design, workflow organization, and delivery of decision-support outputs used by senior stakeholders.

Technical Leadership in Analytics Platform Development. Led renovation of an interactive analytics tool for senior leadership, taking ownership of refactoring, functionality expansion, and stakeholder demonstrations to program directors. Collaborated with engineers and analysts to align tool outputs with mission requirements.

Cross-Functional Mentorship and Client-Facing Leadership. Served as a technical liaison in optimization and supply chain project meeting with clients, guiding onboarding, translating quantitative requirements into engineering specifications, and supporting backend teams during system architecture redesigns.

Engineering Guidance in Applied ML Systems. Provided technical direction across forecasting, optimization, and ML model development efforts, including mentoring junior engineers on ETL processes, API design, and reproducible Python workflows in production environments.